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Fairatmos: Building an AI-powered carbon emissions reduction business with Google Earth Engine and Vertex AI

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GOOGLE CLOUD RESULTS

- Reduced time to demonstrate revenue opportunities from carbon projects from weeks to seconds
- Securely manages a dataset comprising information about 45 million hectares across South East Asia
- Runs multidimensional queries across spatial, government, and public data for rich insights into potential opportunities
- Gained the ability to experiment with and evaluate a range of models

With Earth Engine and Vertex AI, Indonesian sustainability startup Fairatmos provides mapping and revenue opportunity information in seconds to

help farmers and small communities make informed decisions about proceeding with nature-based carbon emission reduction projects.

Climate change is disrupting communities and farmers in countries like Indonesia, yet cost and complexity limits their participation in financially rewarding mitigation programs. Founded in 2022 by Natalia Rialucky Marsudi, the sustainability startup [Fairatmos](#) is revolutionizing access to voluntary carbon markets through AI-powered solutions and cloud-based innovations, utilizing [Earth Engine](#) and [Google Cloud](#) products and services.

According to Fajar Firdaus, Head of Engineering and Machine Learning at Fairatmos, technology is critical to establishing the organization's value proposition and marketing it to potential customers. "It's a new domain for many of them, so we need to communicate our value by demonstrating that our results and analytics are accurate by international standards and government requirements," Firdaus explains. "This enables us to demonstrate the bottom-line improvement they can achieve by adopting a more sustainable model."

To complete the mapping needed to serve its projects, typical carbon project would require remote-sensing specialists to manually digitize maps based on remote-satellite assessment data and classify areas by characteristics such as forest coverage or housing. However, complying with strict regulations governing carbon projects meant development of each map to the required standard could take several weeks. "In each case, this effort would serve only one project in a market rich with potential," says Firdaus.

Fairatmos modernized and accelerated this process, and they defined the requirements of a new solution. "We needed a centralized product that could simplify the process, mitigate the inaccuracies and slow speeds we experienced working locally on laptops without synchronization," says Firdaus. "We also needed support for special data operations and calculations. For example, for our carbon projects, we needed to assess the transformation of forest regions and apply time stamps, time-series data, and other features to provide an accurate, transparent service."

The business determined that working with existing open-source solutions would require it to orchestrate functions and calculations from multiple open-source libraries. "By contrast, Earth Engine had a sound SDK that would enable us to easily develop analysis software for our projects," says Firdaus. "The product also offered a centralized solution we could use to manage the mapping assets that underpinned that analysis, and it natively supported our special data operations and calculations."

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>
> **Fajar Firdaus**
> Head of Engineering, Fairatmos

Seamless integration enables multidimensional queries

Google Cloud's rich AI and data-analytics services also offered Fairatmos opportunities to enhance its modeling and drive internal performance. Finally, with the assessment process incorporating solution providers' environmental credentials, the business received advice from one of its investors that Google Cloud had a comprehensive sustainability program.

Fairatmos opted for Earth Engine, a Google Cloud data architecture centered on the BigQuery analytics data warehouse, and AI services including Vertex AI to execute its transformation. Working with the Google Cloud team and official Google Cloud partner CloudMile, the business transitioned seamlessly to the new cloud platform. “Google Cloud provided strong support on key technical questions, particularly regarding Earth Engine, where the product team advised us on structuring our scripts to deliver scalability,” says Firdaus. “CloudMile helped us optimize pricing and advised us on best practices.”

The business now uses Earth Engine to securely manage a dataset of land-cover classifications spanning 450 million hectares across South East Asia. “This is a vast leap from offline coverage using individual laptops of small projects spanning just a few thousand hectares,” says Firdaus. “With Earth Engine, we’ve removed the need to synchronize project data across laptops, reducing workloads, improving accuracy, and enhancing the scope and quality of our services. For example, the product helps us undertake the core calculations and functions we need to cover the transition from one type of land to another.”

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Fajar Firdaus

Head of Engineering, Fairatmos

The seamless integration between Earth Engine and BigQuery enables Fairatmos to run queries across spatial data and government and public data captured in the warehouse. This combines highly accurate mapping with land-governance data such as tenure, management, and approved usages. The data moved into and stored and processed in BigQuery also enables the business to track teams’ performance against objectives, key results, and metrics set for user activities within its product. Meanwhile, Vertex AI plays a critical role in enhancing the depth and accuracy of Fairatmos’ services. “We use a lot of versions of our internal models for land cover and Vertex AI enables us to record improvements or learning opportunities for each one, giving us the flexibility to experiment easily and cost effectively," says Firdaus.

Fairatmos architecture

Identifying carbon-project revenues in seconds

With Google Cloud and Earth Engine, Fairatmos can help owners of forested land identify the potential benefits of a carbon project in seconds, and closely monitor performance throughout a project. "Our clients tell us how happy they are to receive initial numbers including carbon credit potential and revenue predictions straight away, demonstrating that this improved experience is key to our growth plans," says Firdaus. "The data we provide also includes mapping and other supporting imagery, text, and calculations that identify how our project is more profitable than other options. This helps dispel client uncertainty and builds their confidence in and excitement about carbon projects."

For Firdaus, the operational acceleration and experimentation enabled by Google Cloud is key to the organization's future. "Google Cloud and Earth Engine are helping us deliver superior project executions, which enable us to continue to gain traction in this vitally important market," Firdaus concludes.

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Fairatmos is a climate technology company developing high-quality nature-based carbon offset projects across Southeast Asia. They leverage advanced remote sensing and deep machine learning to transform carbon stock assessments, cutting analysis time from three months to mere seconds while ensuring high accuracy and integrity. Their approach is specifically tailored to the diverse ecosystems of the Southeast Asia region.

Industry: Technology

Location: Indonesia

Products: Earth Engine, BigQuery, Vertex AI

About Google Cloud partner — CloudMile

CloudMile aims to empower enterprises by utilizing AI, machine learning, big data, and Google Cloud services. An official Google Cloud partner, CloudMile has more than 900 clients across Hong Kong, Malaysia, Singapore, and Taiwan.

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